



CCTV cameras in operation

equipment to photograph individuals using CCTV to gather incriminating evidence.

Wired agents and informers.

This technique involves wiring of an agent, informer or a consenting party to overhear, and sometimes record, conversations.

What about individual privacy

Electronic surveillance, when utilised as a tool for national security, law enforcement, industrial espionage or domestic relations, can limit and ultimately negate citizens' right to privacy.

One form of electronic surveillance developed by law enforcement agencies resulted in attaching a bug to a person's telephone line or to a phone booth and recording the person's conversation. Courts have held that this practice constitutes a search under the Fourth Amendment, which protects an individual's privacy rights for situations in which the person has a legitimate expectation of privacy.

Courts have also held that when having a telephone conversation, one would not expect an unknown third-party government agent to listen in on the conversation. A person has a legitimate expectation of privacy if the

person honestly and genuinely believes the location under search to be private and if the reasonable person under same or similar circumstances would believe the location to be private as well.

Therefore law enforcement agencies have more leeway when intercepting communications in a public place than when the interception occurs in a secluded environment. Courts have given law enforcement agencies the freedom to re-

cord conversation during jail visits, provided the monitoring reasonably relates to prison security.

A major argument against the use of these extraordinary means suggests that electronic surveillance violates the very rights it purports to protect.

Failures of electronic surveillance

Electronic surveillance may offer advantages, such as night view using infrared lighting, which recognises what is happening at the place even when it is pitch dark but does not protect against thefts. However, it can deter criminals upfront, especially opportunistic criminals, who are looking for a quick way to steal valuable items. And, if a theft is committed, recordings in the video surveillance can serve as evidence while also enabling identification of the culprit(s).

With state-of-the-art motion detection in surveillance cameras, we can enjoy all-round protection, even in hardly-accessible areas. With the low crime-detection rate for thefts and the high percentage of repeat offenders, advantages of a detailed surveillance with the chance of identifying culprits are pretty obvious.

But even with electronic surveillance, all over the world, thieves, criminals and law breakers are still actively operating in warehouses, production locations and in offices, and are able to design unique methods to avoid electronic surveillance.

For thefts from ATMs, thieves use a device connected to a small laptop and inserted into card acceptance slots on ATMs. Armed with this tool-set, they are able to install malware that is capable of siphoning the customers' card data and personal identification numbers (PINs). The device looks like a green circuit board that is approximately four or five times the length of an ATM card.

Insertion of the circuit board causes the software running on ATMs to crash, temporarily leaving the cash machine with a black, empty screen. Thieves then remove the device and soon after, the machine restarts and begins recording the card and PINs entered by customers who used the compromised machines.

They then return a few days after infecting the ATMs to collect stolen card numbers and PINs. They reinsert the specialised chip card to retrieve purloined data and a separate chip card to destroy evidence of the malware.

Criminals are using improvised electronic devices to electronically unlock vehicles and steal whatever they find inside. These mystery gadgets reportedly recreate the same signals that the key fobs that many of us carry around.

Thieves are using these mystery gadgets to remotely unlock car doors and disable alarm systems. Once a car has been unlocked, it takes these thieves just a few moments to take what they want before leaving without a trace.

As cars get smarter, and more connected, threats also increase. Thanks to all the technology that is in our vehicles these days, these are potentially more vulnerable to hackers than ever. ●